

Supplemental information

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Supplemental information for

The Late Arrival of Domestic Cats in China via the Silk Road after 3,500 Years of Human-Leopard Cats Commensalism

Yu Han^{1,2,3,24}, Songmei Hu^{4,5,24}, Ke Liu^{1,3}, Xiao Xu^{1,3}, Ziyi Li⁶, Sean Doherty⁷, Alexandra Elizabeth Jamieson², Aurelie Manin², Sofia Granja Martins^{2,8,9}, Miaomiao Yang⁵, Chong Yu¹⁰, Juan Wang¹¹, Zhuang Wu¹², Canping Chen¹², Sicheng Han^{1,3}, Daowei Lu^{1,3}, Lanhui Peng^{1,3}, Xianzhu Wu¹³, Wenquan Fan¹⁴, Quanfa Cai¹⁴, Zongliang Cui¹⁵, Jing Yuan¹⁶, Zihan Li^{2,17}, Yang Liu¹⁰, Zhipeng Li¹⁸, Zhendong Liu¹⁸, Qian Ma^{2,19}, Jing Shao⁵, Zhouyong Sun⁵, Fulai Xing⁵, Wuzhan Yang⁵, Shugang Yang¹⁴, Lianjian Yue⁵, Pengcheng Zhang²⁰, Weilin Wang²¹, Huanyuan Zhang-Zheng²², Yan Zhuang^{1,3}, Xin Sun^{1,3}, Yan Pan¹⁹, Xiaohong Wu¹⁹, Laurent A. F. Frantz^{2,23}, He Yu^{1*}, Joel M. Alves^{2,8,9*}, Greger Larson^{2*}, Shu-Jin Luo^{1,3*}

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DOCUMENTS1

This PDF file includes:

Figures S1- S14

Tables S2, S4, and S8

Supplemental references

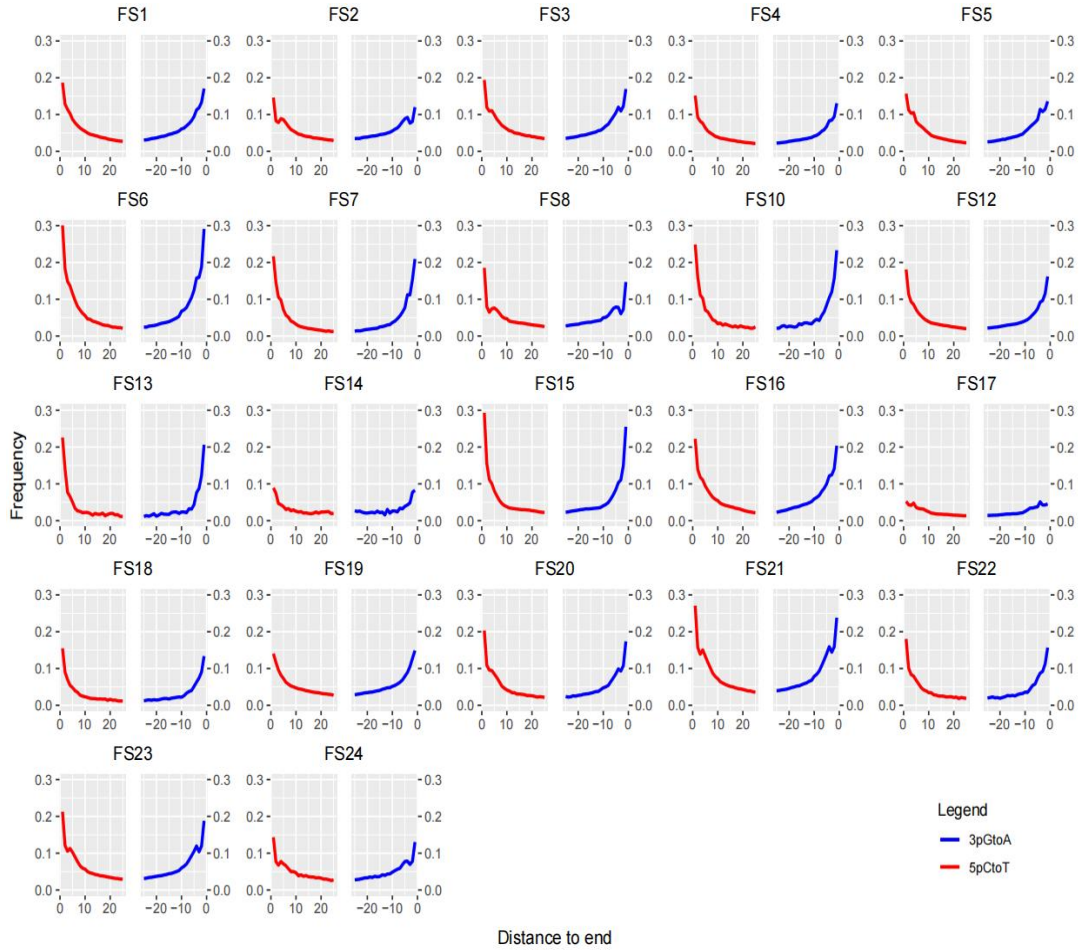


Figure S1. Plots of mapDamage analysis showing damage patterns of 22 ancient Chinese cat specimens used in this study, related to Figures 1-3. The 5' and 3' ends of all samples exhibit a clear deamination pattern signature for ancient DNA.

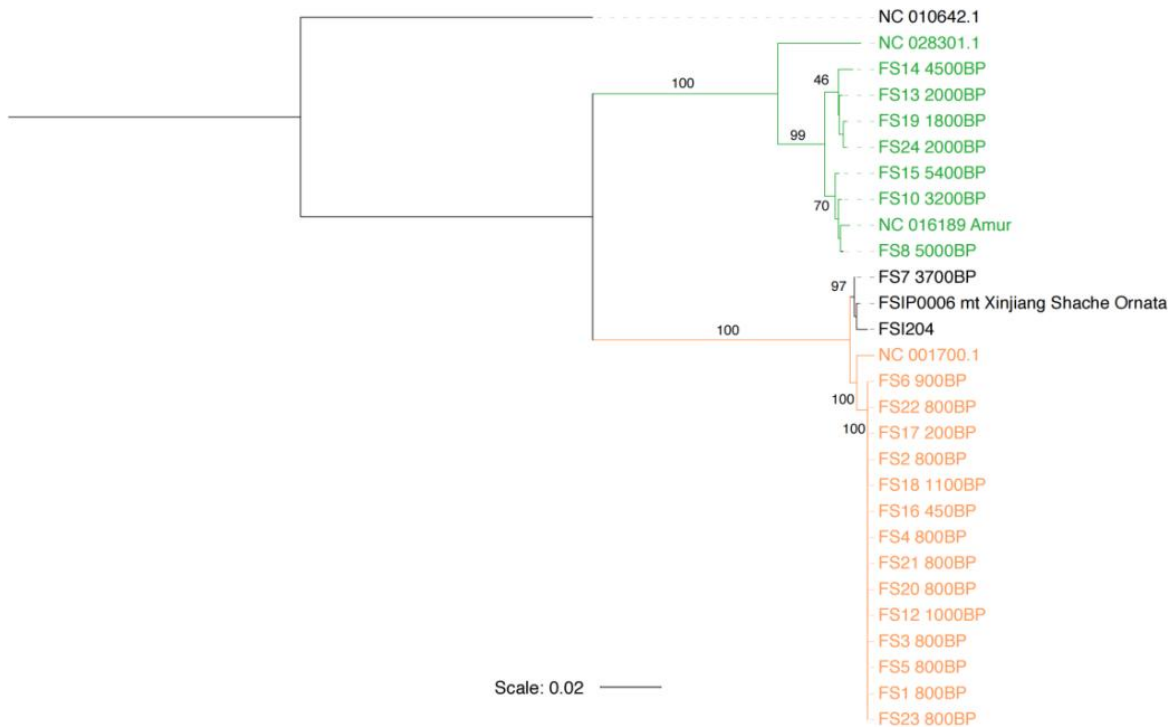


Figure S2. A maximum likelihood (ML) phylogenetic tree constructed based on the mitochondrial genomes of 22 ancient felid samples collected from China, related to Figures 1B and 2.

A TN93+I+G model is implemented with 1,000 bootstrap replicates and a tiger genome (NC_010642.1) is used as an outgroup. The numbers on branches represent the statistical support values based on 1,000 bootstrap replicates.

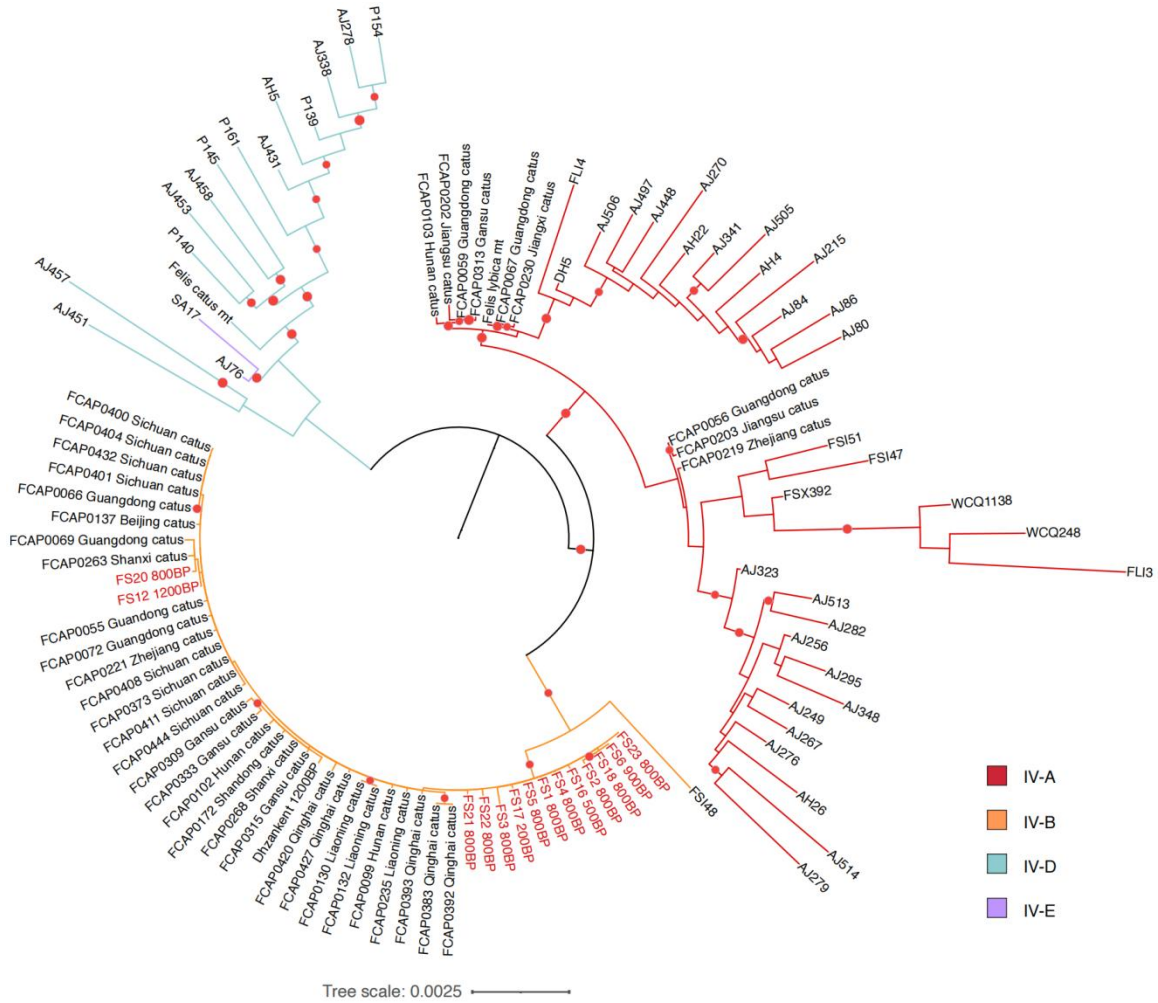


Figure S3. A maximum likelihood (ML) phylogenetic tree based on mitogenomes of ancient and modern domestic cats worldwide, including 14 ancient domestic cats from China, related to Figure 2.

A TN93+I+G model is implemented with 1,000 bootstrap replicates. The missing data rate for Chinese samples is less than 2% each and the sequencing depth is greater than 30. The missing data rate for European or Central Asian samples is generally less than 10% and the sequencing depth is greater than 10. Nodes with significant statistical support ($\geq 70\%$) based on 1,000 bootstrap replicates are labeled with red dots.

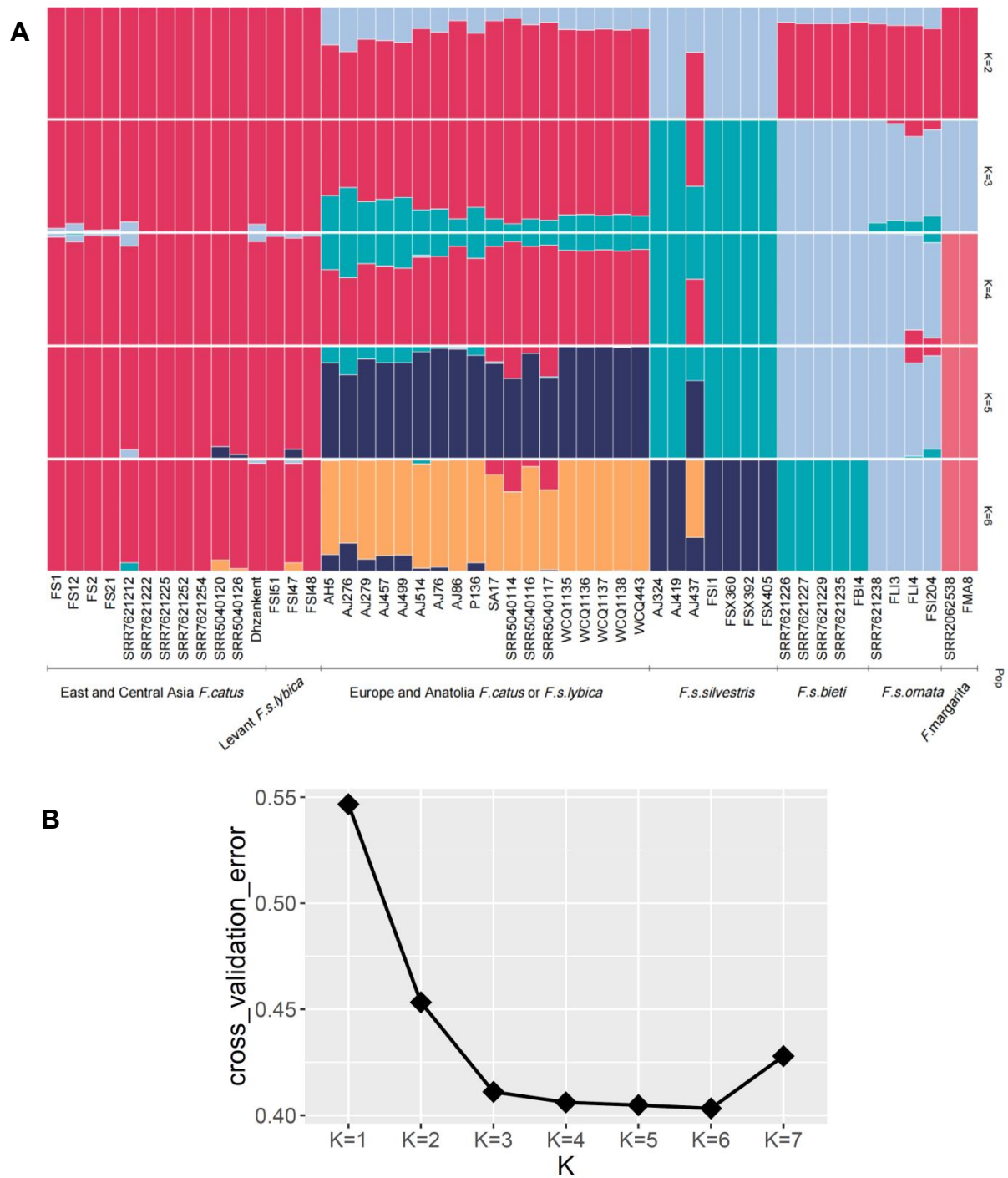


Figure S4. ADMIXTURE analysis for ancient and modern domestic cats and wildcats from Eurasia, related to Figure 3C.

(A) ADMIXTURE clustering results based on the number of assumed ancestral populations (K) from 2 to 6.

(B) Cross-validation (CV) errors for each number of assumed ancestral populations (K) from 1 to 7.

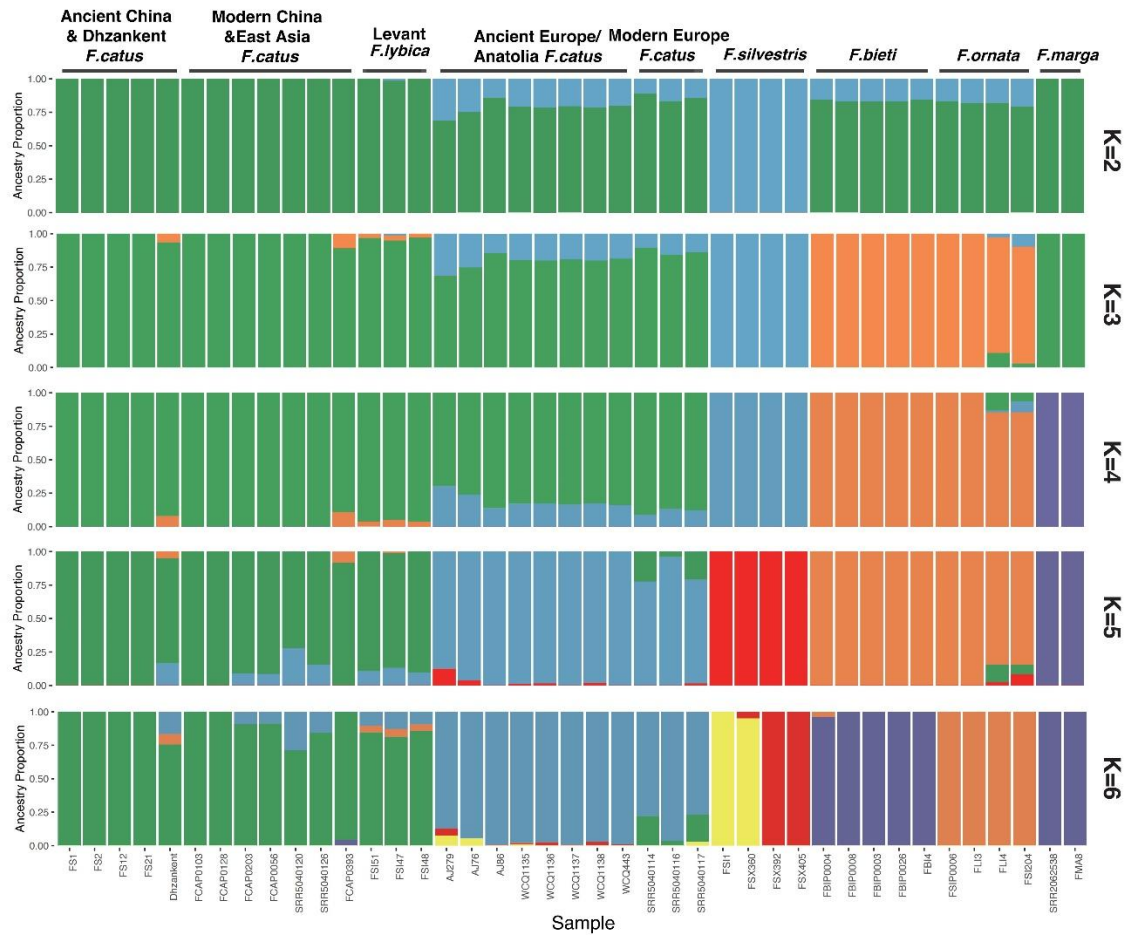


Figure S5. ADMIXTURE analysis including modern samples and ancient individuals with sequencing depth greater than $2\times$, related to Figure 3C. A total of 37 samples are used in this analysis. Pseudo-haploid genotype calling was applied to ancient Chinese samples.

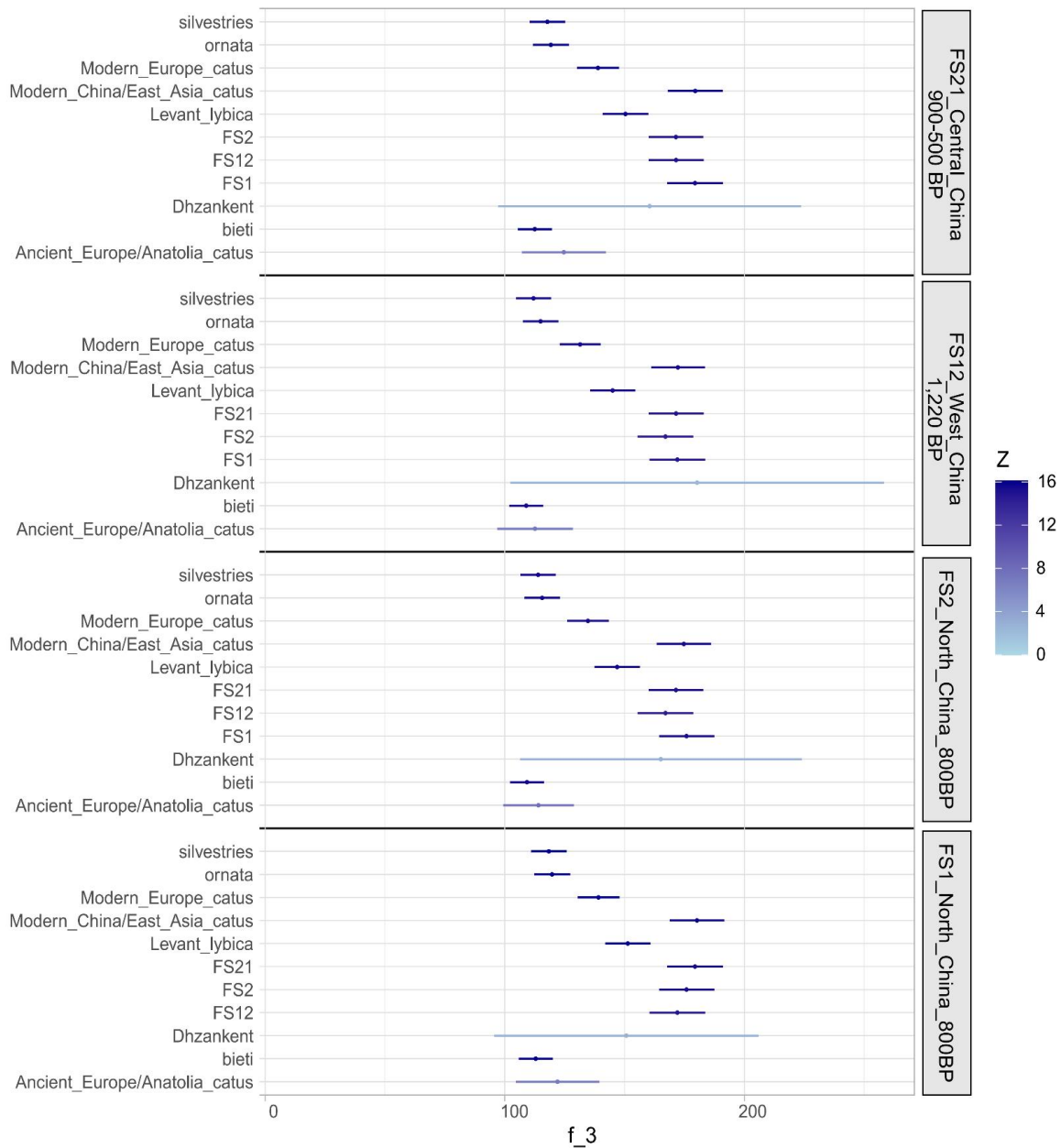


Figure S6. The results of outgroup f_3 -statistics illustrating the affinity between ancient domestic cats from China and other domestic cat populations, related to Figure 3B.

The error bars in the four stacked blocks, from bottom to top correspond to the f_3 values between FS1/FS2/FS12/FS21 and other cat populations (labeled on the right). Bars indicate 1x standard errors (SD). The color gradient, ranging from light to dark blue shows an increased magnitude of the absolute value of Z. This figure, compared to Figure3B, highlights the relationship between ancient domestic cats and modern domestic cats.

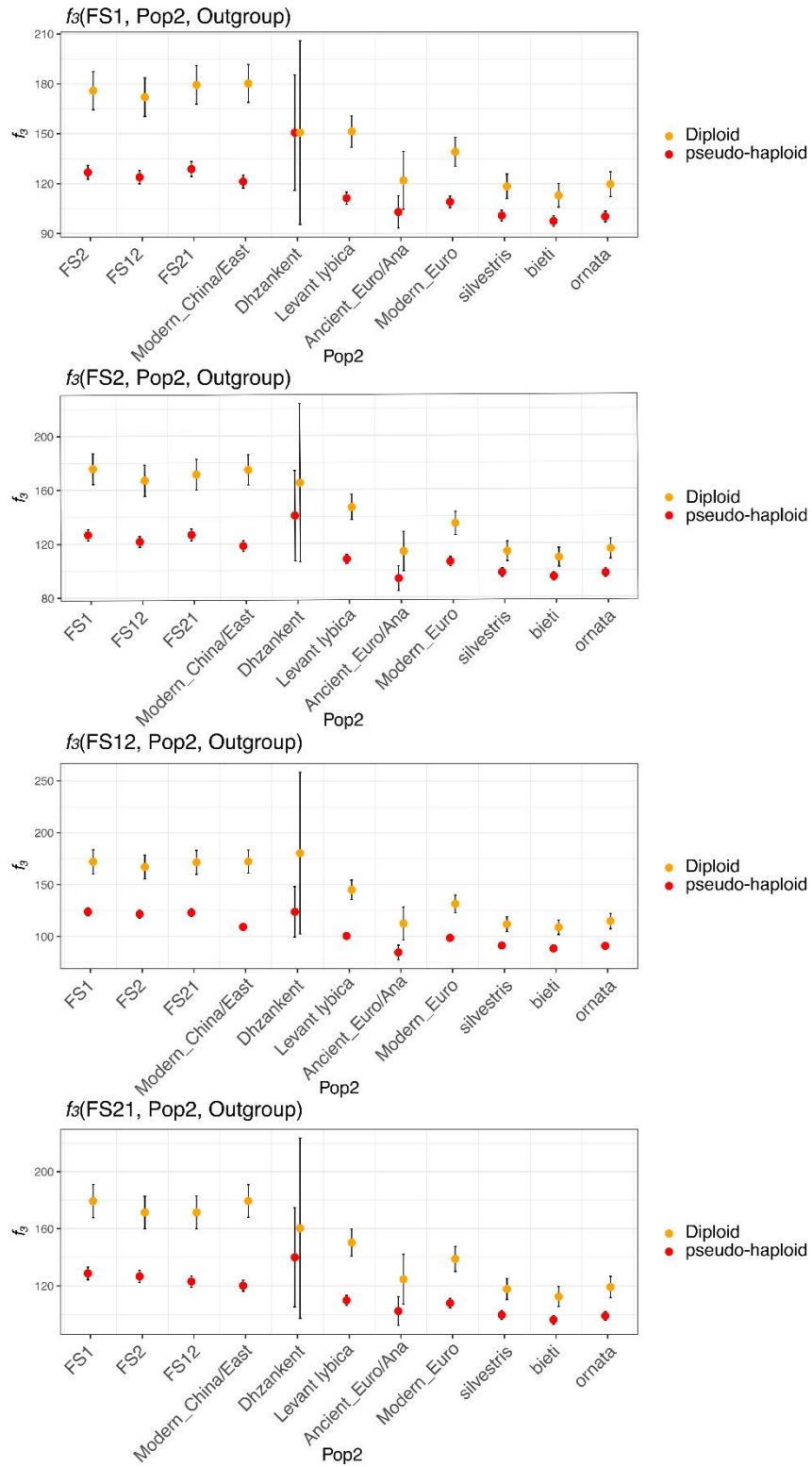


Figure S7. Comparison of outgroup- f_3 statistics based on diploid genotype calling (yellow) and pseudo-haploid genotype calling (red) for ancient Chinese samples, related to Figure 3B.

Error bars indicate 1x standard error.

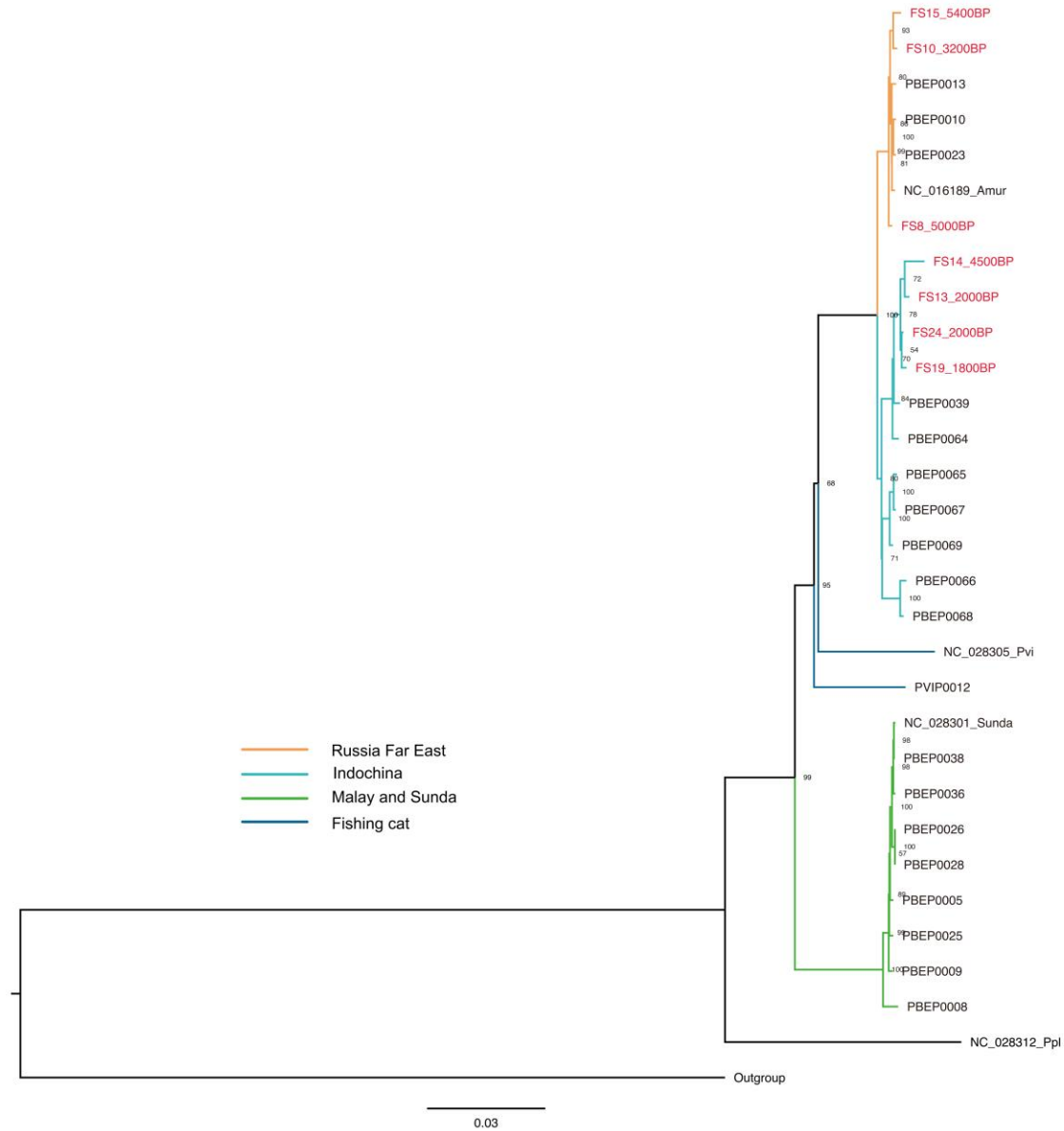


Figure S8. A maximum likelihood (ML) phylogenetic tree based on leopard cat mitochondrial genomes, related to Figures 1A and 1B.

The tree is reconstructed based on ancient leopard cats from the middle Yellow River Basin of China (n=7) and modern leopard cats from the Russian Far East (n=4), Indochina (n=6), Malay Peninsula (n=6), and Sundaland (n=4). Two fishing cats (*P. viverrinus*) and a flat-headed cat (*P. planiceps*) are included in the analysis. The tiger reference genome (NC_010642.1) is used as an outgroup. The sample names of ancient leopard cats from the Yellow River Basin are marked in red.

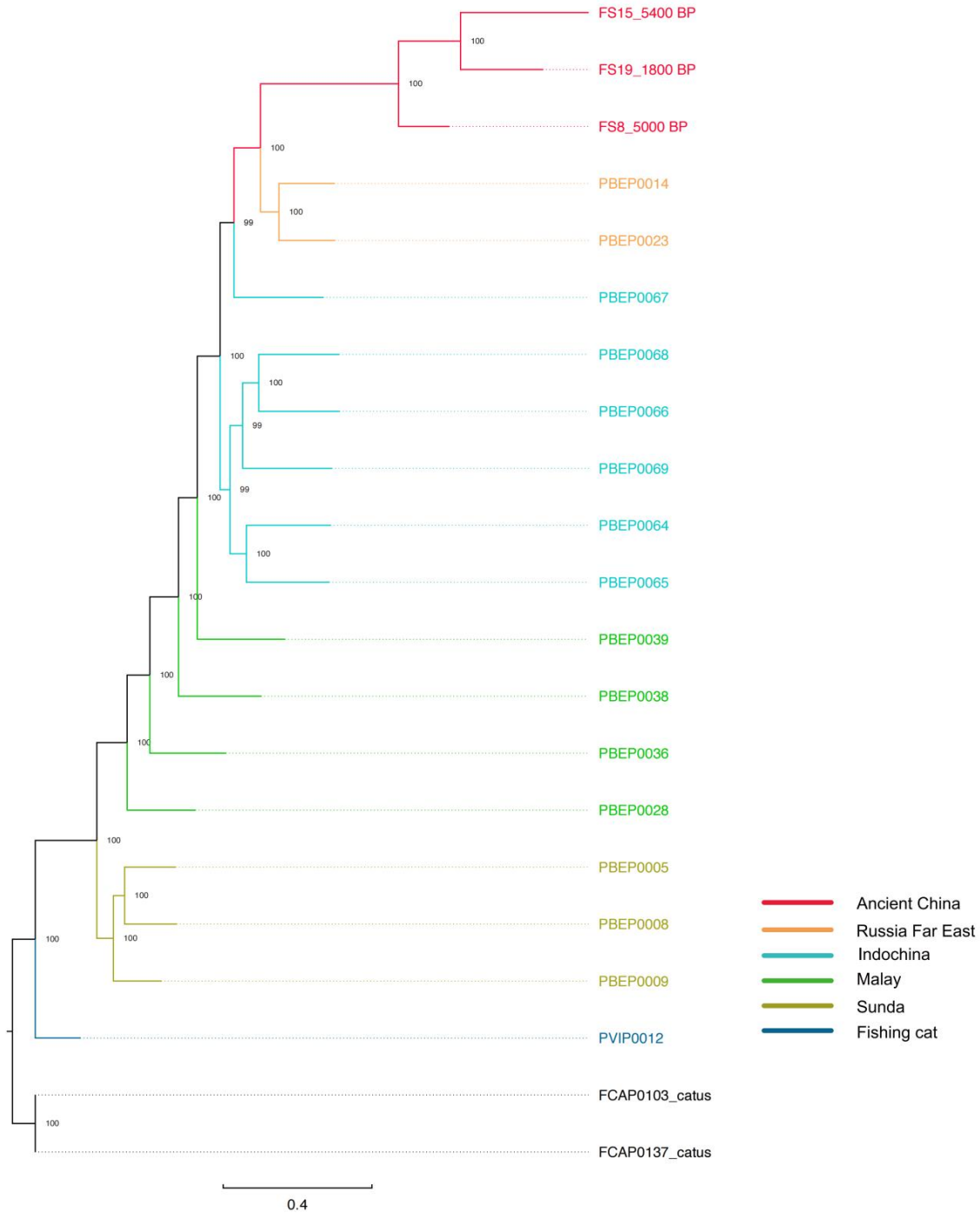


Figure S9. A maximum likelihood (ML) phylogenetic tree based on ancient and modern leopard cats, related to Figures 1A and 1B.

The tree is reconstructed based on autosomal SNPs from ancient specimens from China (n=3) and modern samples from the Russian Far East (n=2), Indochina (n=6), Malay Peninsula (n=4), and Sunda (n=3). A fishing cat (*P. viverrinus*) is included in the analysis.

Two domestic cats are used as outgroups. Numbers at each node represent the statistical support values based on 200 bootstrap replicates.

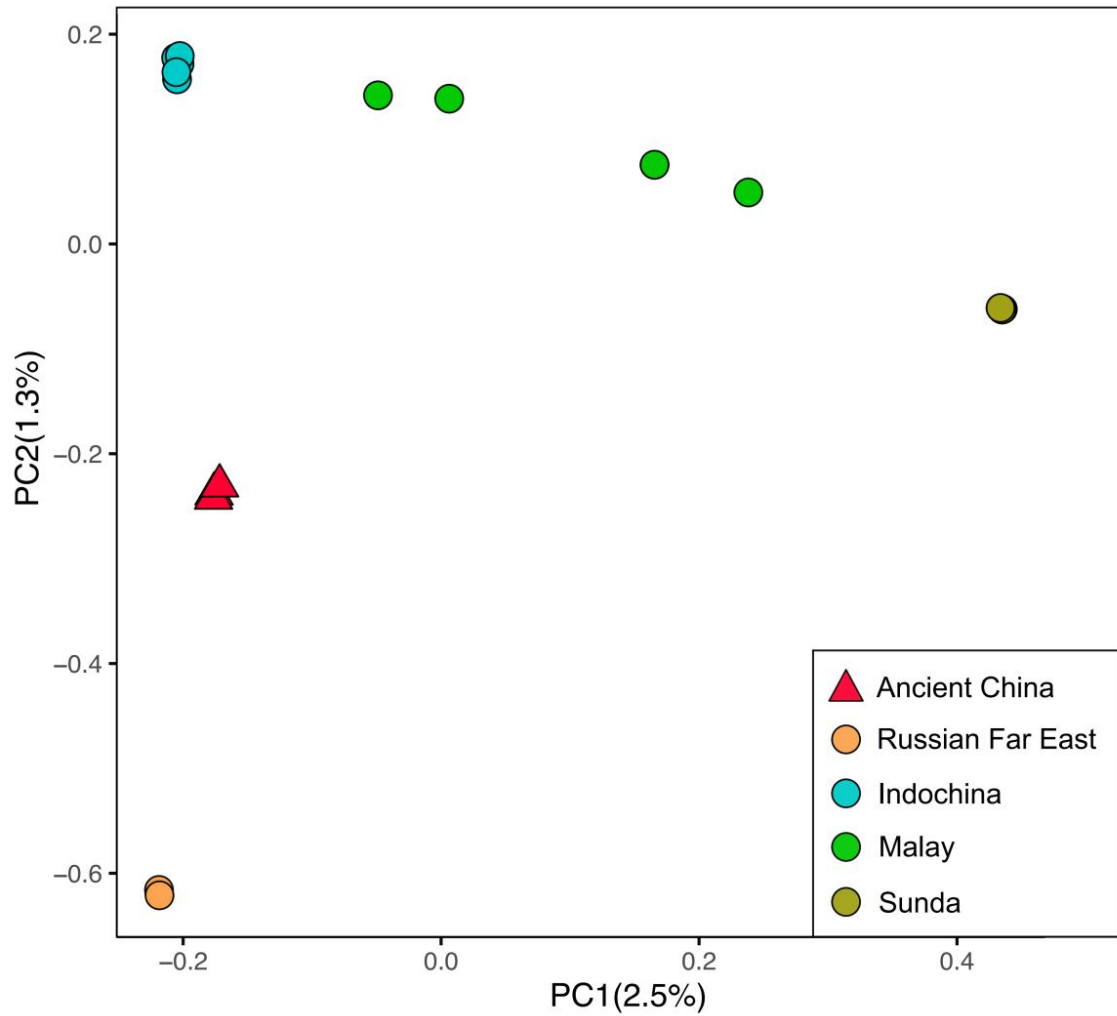


Figure S10. Principal component analysis (PCA) of leopard cats, related to Figure 1A.

PCA is first conducted based on autosomal SNPs from high-quality genome data from modern leopard cats. Data from ancient leopard cats are then projected onto it. Three ancient leopard cats from China and 15 modern leopard cats from range-wide, including the Russian Far East (n=2), Indochina (n=6), Malay Peninsula (n=4), and Sundaland (n=3), are included in the analysis.

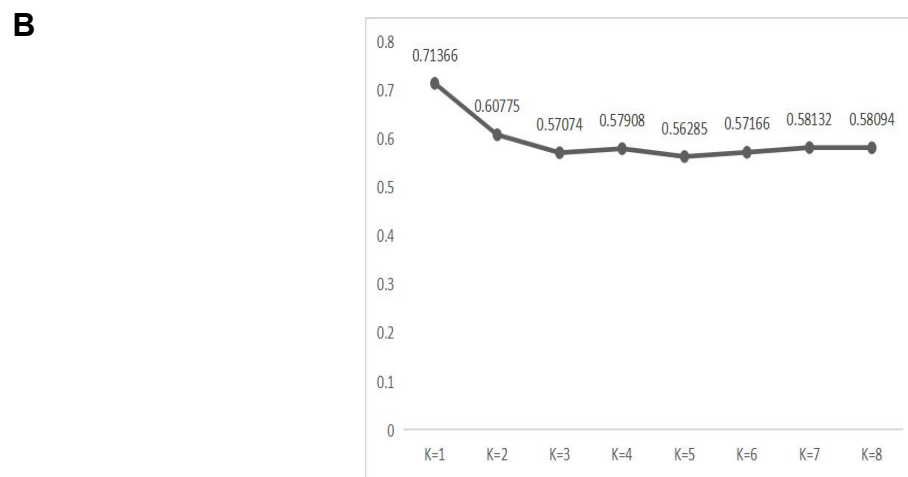
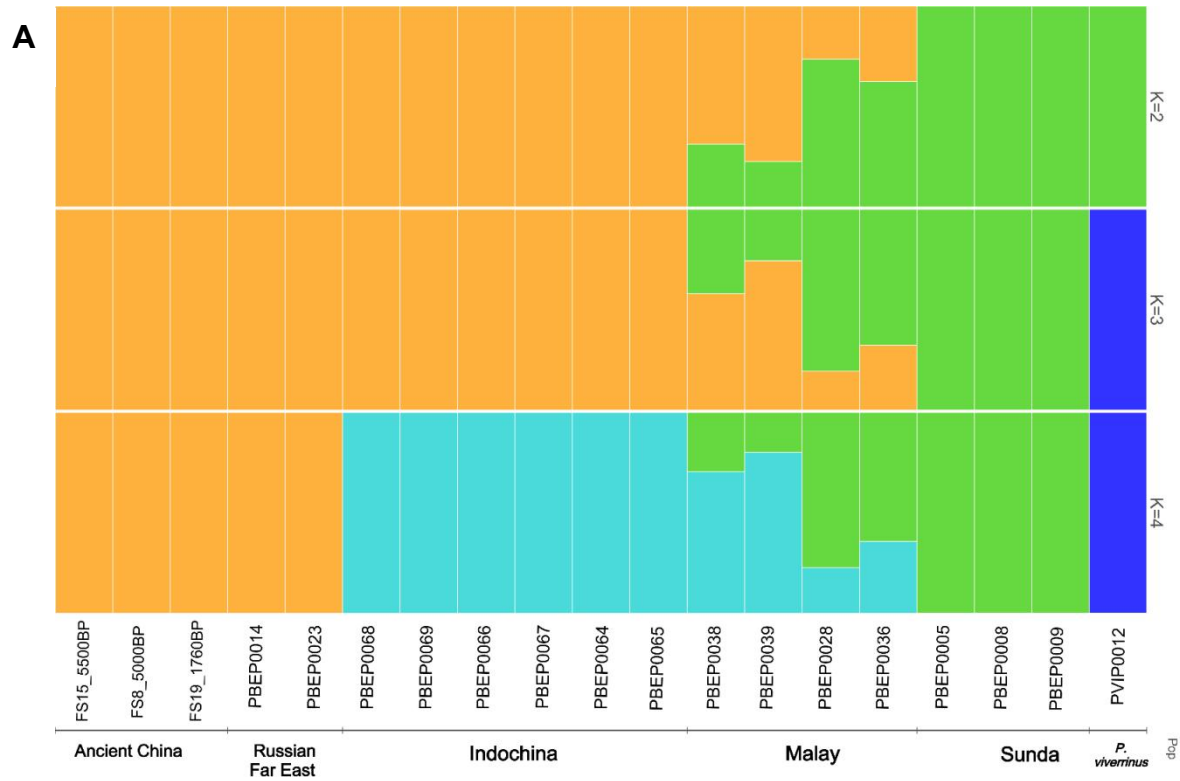


Figure S11. ADMIXTURE analysis of leopard cats, related to Figure 1A.

(A) Clustering results based on the number of assumed ancestral populations (K) from 2 to 4. A total of 18 specimens are used in the analysis, including ancient leopard cats from China (n=3) and modern leopard cats from the Russian Far East (n=2), Indochina (n=6), Malay (n=4), and Sundaland (n=3). A fishing cat was used as an outgroup.

(B) Cross-validation (CV) errors for each number of assumed ancestral populations (K) from 1 to 8.

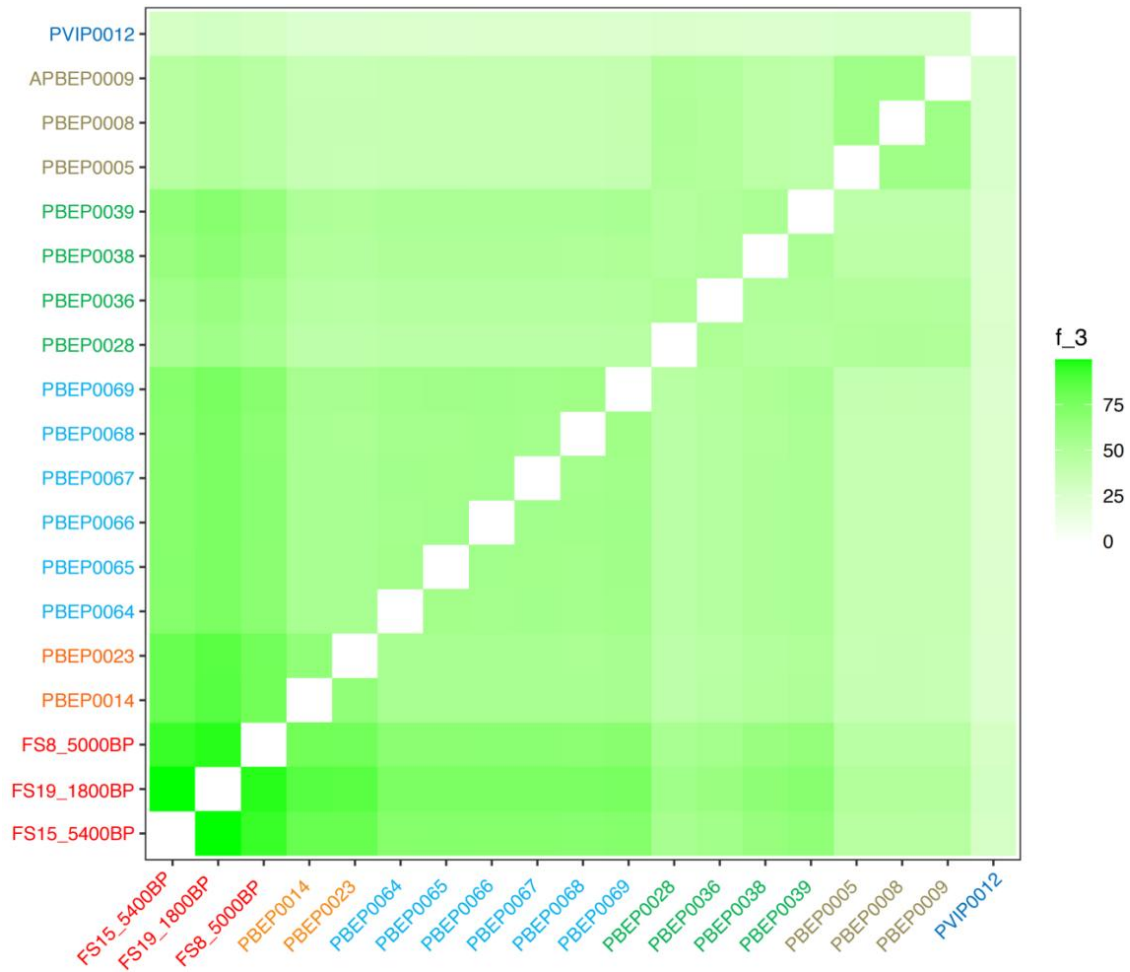


Figure S12. Outgroup f_3 -statistics analysis of leopard cats, related to Figure 1A. A heatmap of the outgroup f_3 -statistics analysis. Two domestic cats are used as outgroups. The shared drifts of different pairs of individuals of leopard cats or fishing cats are tested. The color gradient, ranging from light to dark green, shows the increased magnitude of f_3 values. The sample names from different populations were labeled with different colors. Ancient Chinese leopard cats (n=2) were marked in red; Russian Far East (n=2) in orange; Indochina (n=6) in light blue; Malay (n=4) in green; Sunda (n=3) in brownish yellow; and fishing cat (n=1) in dark blue.

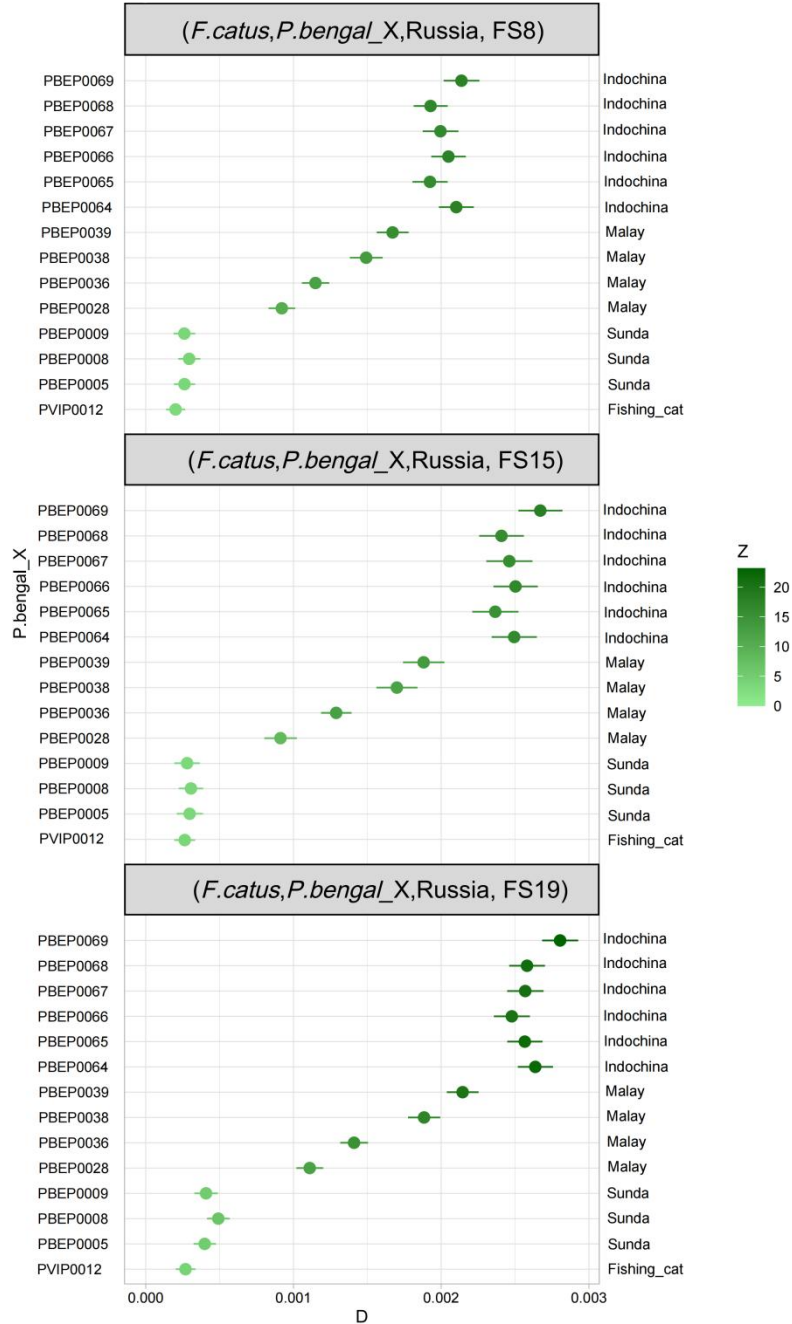


Figure S13. D-statistics analysis of leopard cats, related to Figure 1A.

The population combinations were set as (Outgroup, *P. bengalensis* X, Russian Far East, ancient China) (labeled on the top of each block). Two domestic cats were used as outgroups. Leopard cats from Indochina, Malay Peninsula, Sundaland, and a fishing cat are individually placed in *P. bengalensis* X to evaluate the signals of gene flow between the target ancient Chinese leopard cat and a modern leopard cat from *P. bengalensis* X, relative to a Russian Far East leopard cat. The vertical axis shows the ID of leopard cats in *P. bengalensis* X (on the left) and their population assignments (on the right). The horizontal axis shows the D scores with 1x standard errors (SD). The color gradient, ranging from light to dark green, indicates an increased magnitude of $|Z|$ values.

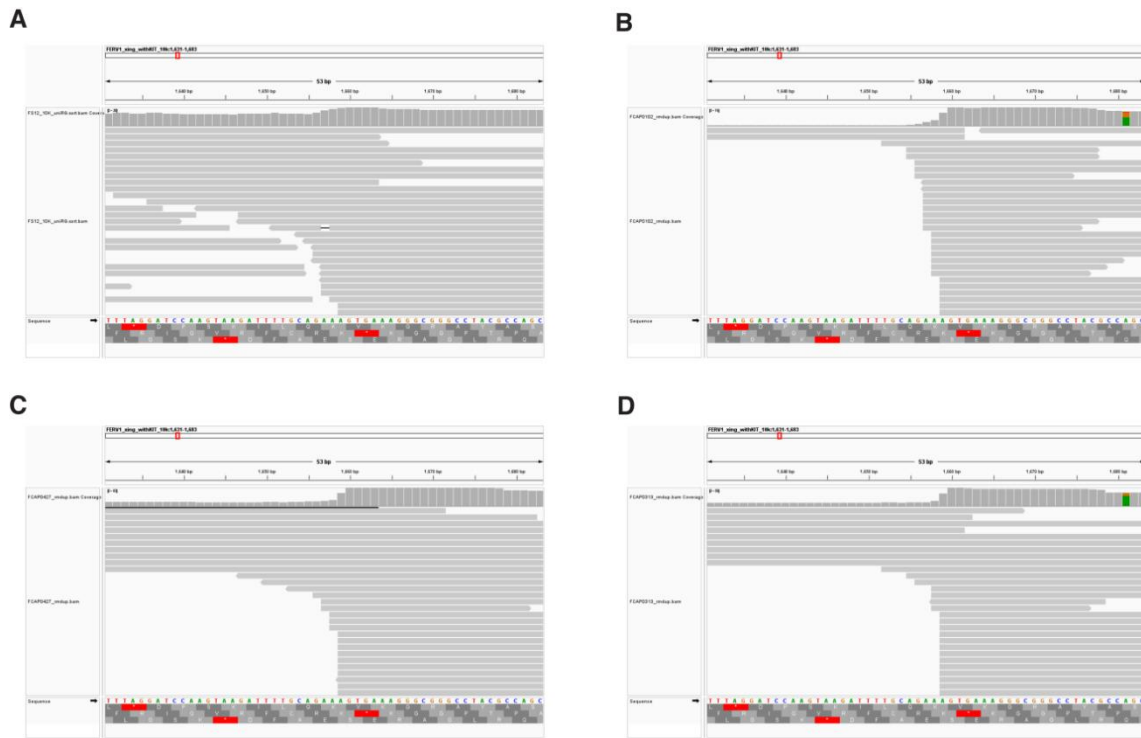


Figure S14. Mapping patterns of the sequencing data to the domestic cat's *KIT* gene with *FERV1* insertion, related to Figure 4A.

- (A) Alignment of FS12's *KIT* gene with *FERV1* insertion.
- (B) Alignment of a wild-type domestic cat *KIT* gene without *FERV1* insertion.
- (C) Alignment of a domestic cat homozygous for *FERV1* insertion.
- (D) Alignment of a domestic cat heterozygous for *FERV1* insertion.

Table S2. Radiocarbon dating of small cat specimens from archeological sites of China, related to Figure 1C

ID	Site	Location	Context date	Lab IDs	Uncalibrated date (BP)	Date cal. BCE/CE (95.4%, overall range IntCal 20)
FS12	Tongwan City	Western China	Song Dynasty	BA212540	1220+/-20	706 -728 cal CE (7.5%) (1244 -1222 cal BP) 772 - 883 cal CE (88.0%) (1178 -1067 cal BP)
FS13	Yangling Tomb	Central China	Han Dynasty	BA212541	2095+/-20	168 - 48 cal BCE (95.4%) (2118 - 1998 cal BP)
FS16	Zhengzhou Shang City	Central China	Tang Dynasty	Beta - 594436	450+/-30	1413 - 1480 cal CE (95.4%) (537 - 470 cal BP)
FS17	Shanliangzi	Northern China	Han Dynasty	Beta - 594435	140+/-30	1797 - 1944 cal CE (153 - 6 cal BP) (58.2%); 1671 - 1779 cal CE (37.2%) (279 - 171 cal BP)
FS19	Zheng and Han City	Central China	Han Dynasty	Beta - 594437	1760+/-30	200 - 380 cal CE (95.4%) (1750 - 1570 cal BP)
FS24	Han Chang'an City	Central China	Han Dynasty	BA212542	1995+/-20	44 cal BCE - 76 cal CE (95.4%) (1994 -1874 cal BP)

Notes: Six samples were submitted for accelerator mass spectrometry (AMS) dating. Three samples (FS16, FS17, and FS19) were dated at Beta Analytic in the United States, while the remaining samples (FS12, FS13, and FS24) were processed at the School of Archaeology, Peking University, China. All radiocarbon dates reported were calibrated using the INTCAL20 calibration curve¹.

Table S4. *D*-statistics of leopard cats from modern Asia and ancient China, related to Figures 1A, 1B and S13

Outgroup	Pop1	Pop2	Pop3	D	std.err	Z	SNPs
<i>Felis catus</i>	PBEP0068	Russian Far East	FS8	0.001928	0.000114	16.92	2089721
<i>Felis catus</i>	PBEP0069	Russian Far East	FS8	0.002137	0.00012	17.775	2086193
<i>Felis catus</i>	PBEP0066	Russian Far East	FS8	0.002049	0.000116	17.679	2082559
<i>Felis catus</i>	PBEP0067	Russian Far East	FS8	0.001995	0.00012	16.601	2087328
<i>Felis catus</i>	PBEP0064	Russian Far East	FS8	0.002102	0.000117	17.903	2095996
<i>Felis catus</i>	PBEP0065	Russian Far East	FS8	0.001924	0.000118	16.301	2081063
<i>Felis catus</i>	PBEP0038	Russian Far East	FS8	0.001492	0.00011	13.516	2090739
<i>Felis catus</i>	PBEP0039	Russian Far East	FS8	0.001671	0.000107	15.668	2089949
<i>Felis catus</i>	PBEP0028	Russian Far East	FS8	0.000921	0.000089	10.387	2089592
<i>Felis catus</i>	PBEP0036	Russian Far East	FS8	0.001149	0.000092	12.492	2090655
<i>Felis catus</i>	PBEP0005	Russian Far East	FS8	0.000262	0.000071	3.703	2096009
<i>Felis catus</i>	PBEP0008	Russian Far East	FS8	0.000294	0.000074	3.954	2090335
<i>Felis catus</i>	PBEP0009	Russian Far East	FS8	0.000261	0.000073	3.573	2090861
<i>Felis catus</i>	PVIP0012	Russian Far East	FS8	0.000202	0.000064	3.173	2085955
<i>Felis catus</i>	PBEP0068	Russian Far East	FS15	0.002408	0.00015	16.032	2335310
<i>Felis catus</i>	PBEP0069	Russian Far East	FS15	0.002671	0.000148	18.103	2331886
<i>Felis catus</i>	PBEP0066	Russian Far East	FS15	0.002503	0.000149	16.786	2328008
<i>Felis catus</i>	PBEP0067	Russian Far East	FS15	0.002461	0.000155	15.885	2333039
<i>Felis catus</i>	PBEP0064	Russian Far East	FS15	0.002494	0.000152	16.439	2341941
<i>Felis catus</i>	PBEP0065	Russian Far East	FS15	0.002366	0.000155	15.312	2326744
<i>Felis catus</i>	PBEP0038	Russian Far East	FS15	0.0017	0.000138	12.338	2336410
<i>Felis catus</i>	PBEP0039	Russian Far East	FS15	0.001881	0.000139	13.565	2335622
<i>Felis catus</i>	PBEP0028	Russian Far East	FS15	0.000912	0.000109	8.337	2335303
<i>Felis catus</i>	PBEP0036	Russian Far East	FS15	0.001289	0.000102	12.622	2336387
<i>Felis catus</i>	PBEP0005	Russian Far East	FS15	0.000297	0.000087	3.432	2342306
<i>Felis catus</i>	PBEP0008	Russian Far East	FS15	0.000306	0.000083	3.709	2336127
<i>Felis catus</i>	PBEP0009	Russian Far East	FS15	0.00028	0.000086	3.264	2336820
<i>Felis catus</i>	PVIP0012	Russian Far East	FS15	0.000264	0.000071	3.728	2331636
<i>Felis catus</i>	PBEP0068	Russian Far East	FS19	0.002581	0.00012	21.513	2613447
<i>Felis catus</i>	PBEP0069	Russian Far East	FS19	0.002805	0.000121	23.159	2608527
<i>Felis catus</i>	PBEP0066	Russian Far East	FS19	0.002478	0.000121	20.545	2603553
<i>Felis catus</i>	PBEP0067	Russian Far East	FS19	0.002569	0.000122	21.055	2609780
<i>Felis catus</i>	PBEP0064	Russian Far East	FS19	0.002637	0.000118	22.31	2621480
<i>Felis catus</i>	PBEP0065	Russian Far East	FS19	0.002566	0.000118	21.719	2600836
<i>Felis catus</i>	PBEP0038	Russian Far East	FS19	0.001884	0.000108	17.379	2614086
<i>Felis catus</i>	PBEP0039	Russian Far East	FS19	0.002145	0.000107	20.029	2612845
<i>Felis catus</i>	PBEP0028	Russian Far East	FS19	0.00111	0.00009	12.295	2612407
<i>Felis catus</i>	PBEP0036	Russian Far East	FS19	0.00141	0.000093	15.156	2614254
<i>Felis catus</i>	PBEP0005	Russian Far East	FS19	0.000399	0.000075	5.324	2621116
<i>Felis catus</i>	PBEP0008	Russian Far East	FS19	0.000491	0.000076	6.437	2613456
<i>Felis catus</i>	PBEP0009	Russian Far East	FS19	0.000408	0.000078	5.198	2614325
<i>Felis catus</i>	PVIP0012	Russian Far East	FS19	0.000269	0.000066	4.061	2607464

Table S8. Information of ancient paintings from China involving images of domestic cats within the time frame of this study from the Tang Dynasty to modern time (827 - 1930 CE), related to STAR Methods

Theme	Date	Period	Type	Black	White	Unclear	Number of individuals	Collector/Source
<i>M126 of the Tombs of the Tang and Song Dynasties at Liujiazhuang Locus North in Anyang City, Henan</i>	827 CE	Tang	Tomb mural	1	1	2	3	The excavation report f the Tombs of the Tang and Song Dynasties at Liujiazhuang Locus North in Anyang City, Henan
<i>Tang Dynasty Mural Tomb at Beiguan, Anyang City, Henan</i>	835 - 907 CE	Tang	Tomb mural	1	1	0	1	Preliminary Report on the Excavation of Tang Dynasty Mural Tombs at Beiguan, Anyang City, Henan
<i>Anyang Xiaonanhai Mural</i>	960 - 1127 CE	Song	Tomb mural	1	0	0	1	Zhaoling's Painted Brick Chamber from the Tang Dynasty
<i>Dengfeng Heishangou Song Tomb</i>	11th -12th centuries CE	Song	Tomb mural	1	1	0	1	Research on Images from Song Dynasty Tomb in Dengfeng's Heishan Gou by Yi Qing
<i>Xu Congbin's Tomb</i>	982 CE	Song	Tomb mural	1	1	0	1	Newly Added Fort's Liao Dynasty Tomb of Xu Congyuan
<i>Chen Shu's Cat and butterfly Painting</i>	17th -18th centuries CE	Qing	Legacy painting	1	2	0	2	The Palace Museum in Beijing
<i>Cheng Zhang's Two Cats Peering at Fish</i>	1930 CE	modern times	Legacy painting	2	2	0	2	The Palace Museum in Beijing
<i>Qiu Ying's Portrait of a Lady</i>	16th century CE	Ming	Legacy painting	2	3	0	3	The Palace Museum in Beijing
<i>Li Di's Winter Play with Infant</i>	10th - 11th centuries CE	Song	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Wealthy FloralCat Painting</i>	12th - 13th centuries CE	Song	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Siblings in Unity Painting</i>	13th - 14th centuries CE	Yuan	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Yi Yuanji's Cat and Monkey</i>	11th century CE	Song	Legacy painting	2	2	0	2	National Palace Museum in Taipei
<i>Wealth in the Pot Painting</i>	15th century CE	Ming	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Cat Under the Banana Shade</i>	14th - 17th centuries CE	Min	Legacy painting	4	4	0	4	The Metropolitan Museum of Art
<i>Xuanzong's Scroll of FiveCats</i>	15th century CE	Min	Legacy painting	4	5	0	5	The Metropolitan Museum of Art
<i>Tao Cheng'sCat in Fragrant Grass</i>	15th century CE	Min	Legacy painting	6	5	0	14	National Palace Museum in Taipei
<i>Li Di'sCat and Dragonfly</i>	10th - 11th centuries CE	Song	Legacy painting	1	1	0	1	Collection of Osaka City Museum of Fine Arts, Japan
<i>Cat Painting</i>	10th - 11th centuries CE	Song	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>LittleCat Shadow</i>	10th - 11th centuries CE	Song	Legacy painting	0	1	0	1	National Palace Museum in Taipei
<i>Cat and Infant Play Painting</i>	10th - 13th centuries CE	Song	Legacy painting	2	2	0	2	Museum of Fine Arts, Boston
<i>Song Huizong's Cat and butterfly Painting (Signature)</i>	12th centuries CE	Song	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Li Di's Okra Mountain and Rock</i>	13th century CE	Southern Song	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Shen Zhenlin's Cat and butterfly in Spring Album, WisteriaCat</i>	19th century CE	Qing	Legacy painting	10	15	0	15	National Palace Museum in Taipei
<i>Shang Xi's Playful Cat Scroll</i>	15th century CE	Ming	Legacy painting	1	2	0	2	National Palace Museum in Taipei
<i>Five Cats Painting</i>	12th - 13th centuries CE	Song	Legacy painting	4	4	0	5	Private Collection
<i>Shen Zhenlin's Cat and Bamboo</i>	19th century CE	Qing	Legacy painting	1	3	0	3	National Palace Museum in Taipei
<i>Zhou Wenju's Portrait of a Lady</i>	10th century CE	Five Dynasties	Legacy painting	1	1	0	1	National Palace Museum in Taipei
<i>Liangkai's Scroll of Leisurely Delights of the cats</i>	12th century CE	Song	Legacy painting	2	5	0	5	Nara Yamato Bunkakan in Japan
<i>Luo Zonggui's Hibiscus Cat Play Fan Painting</i>	10th - 13th centuries CE	Song	Legacy painting	1	2	0	2	N/A
<i>Mao Yi's Sichuan Hibiscus and Wandering Cat</i>	13th century CE	Southern Song	Legacy painting	3	3	0	4	Nara Yamato Bunkakan in Japan
<i>Jin Qing's Twin Cats</i>	10th - 13th centuries CE	Song	Legacy painting	1	2	0	2	N/A
<i>Wang Weilie's Cat and Rock</i>	17th century CE	Qing	Legacy painting	1	1	0	1	National Art Museum of China
<i>Wang Guxiang's Cat and butterfly Painting</i>	16th century CE	Ming	Legacy painting	1	1	0	1	N/A
				62	77	2	91	
				68%	85%			

Supplemental references

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